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The effects of initial substrate concentration, C/N ratio, and temperature on solid-state anaerobic digestion from composting rice straw

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HIGHLIGHTS

- Solid-state anaerobic digestion from composting rice straw was studied.
- Microbial community was determined by the high-throughput sequencing technique.
- The optimal condition for biogasifcation is 35.6 °C, 20% ISC and C/N of 29.6:1.
- Methanogen was shifted from *Methanobacterium* to *Methanoculleus* during the process.

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ABSTRACT

This study investigated the possibilities of improving the biogasification from solid-state anaerobic digestion (SS-AD) of composting rice straw (RS) based on the optimized digestion temperature, initial substrate concentration (ISC) and C/N ratio. RS compounds, such as lignin, cellulose, and hemicellulose, were significantly degraded after composting. A significant interactive effect of temperature, ISC and C/N ratio was found on the biogasification of SS-AD of composting RS, and a maximum biogas production was achieved at 35.6 °C, with a 20% ISC and a C/N ratio of 29.6:1. The verification experiment confirmed the optimization results. High-throughput sequencing analysis indicated that microbial communities in the SS-AD mainly consist of *Methanobacteria*, *Bacteroidia*, *Clostridia*, *Betaproteobacteria*, and *Gammaproteobacteria*. A dominant *Methanobacteria* was shifted from *Methanobacterium* to *Methanoculleus* during the SS-AD process. This study provides novel information about the interdependent effects and microbial behavior of AD.

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1. Introduction

As a result of global warming, increased waste disposal and energy costs and the need for environmentally sustainable waste management, growing attention has been devoted to the bioconversion of biomass into methane-rich biogas. This process demonstrates a wide range of beneficial effects on the environment, energy and human health, especially in rural areas of developing countries (Khoramnia et al., 2011). China is one of the largest agricultural countries and produces over 600 million tons of crop straw residues every year, ranking first in the world (MOA, 2011). Among these sources, rice straw (RS) represents the largest portion and accounts for 32.3% of the total crop straw output (MOA, 2011). Although a few traditional methods of RS disposal or utilization exist, such as providing energy for household cooking and heating, most of the crops are burned in a field, resulting in the emission of severe environmental pollution, such as greenhouse gases and nitrogen oxide (Yang et al., 2008). If the RS can be efficiently transformed into biogas, then the energy shortage in rural areas of China, as well as CO_2 emission problems, will be alleviated to a certain extent (Sun et al., 2005).

Anaerobic digestion (AD) has been successfully implemented in biomass conversion due to its capability of reducing chemical oxygen demand and biological oxygen demand from organic wastes while providing clean energy (Chen et al., 2008). The AD process can operate in both liquid and solid states in terms of total solid (TS) content. In general, the TS content of liquid AD (L-AD) systems ranges from 0.5% to 15%, whereas solid-state AD (SS-AD) systems usually operate at TS contents of higher than 15%. Compared with L-AD, SS-AD has advantages partially due to smaller reactor

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capacity requirements, less energy used for heating, and no processing energy needed for stirring. The digestate of SS-AD can be used as fertilizer or pelletized fuel due to the low moisture content, which is much easier to handle than the effluent remaining after liquid AD. Additionally, problems related to the floating and stratification of fibrous material in L-AD does not occur in SS-AD (Chanakya et al., 1993), and the biogas production from SS-AD is comparable to the output of L-AD (Luning et al., 2003). These advantages resulted in an early development of SS-AD systems at the industrial scale. However, because of a lack of scientific knowledge, most digesters based on SS-AD processes are usually dependent on the empirical knowledge of the operators; the AD performances are not optimized and could therefore be improved. Several process parameters can be optimized to operate SS-AD: initial substrate concentration (ISC), carbon/nitrogen (C/N) ratio, physico-chemical pretreatment of the substrate, type of feedstock, and temperature (Karthikeyan and Visvanathan, 2012).

A primary parameter that drives SS-AD is the ISC (i.e., water content). SS-AD industrial plants are operated from 20% to 30% TS and up to 40% TS for certain technologies (Karthikeyan and Visvanathan, 2012). Increasing the TS content leads to lower process performance in terms of methane production and substrate conversion (Abbassi-Guendouz et al., 2012). Research by Fernandez et al. (2008) showed that the methane yield at 30% solids content was 17% less than that at 20%. Le Hyaric et al. (2012) observed that TS content affects the various steps of the AD process: hydrolysis, acidogenesis, acetogenesis and methanogenesis.

Digestion temperature is another key factor affecting the performance of SS-AD, both in batch and continuous mode. Several authors investigated the effect of temperature on the SS-AD process and obtained different results. The findings of Mashad et al. (2004) indicated that SS-AD under mesophilic conditions exhibited a poor start-up performance, whereas operating SS-AD systems under thermophilic conditions (55.8 °C) can accelerate the AD process. Kim et al. (2002) found a similar phenomenon of higher biogas yield during anaerobic digestion of organic fraction of municipal solid waste (OFMSW) at thermophilic conditions than that at mesophilic conditions. However, recent research by Li et al. (2010) found that total methane production increased with temperature from ambient to mesophilic conditions, whereas acidification occurred in thermophilic conditions and led to the inhibition of biogas production. Clearly, the operation temperature for the SS-AD is currently controversial. From the economical view, the mesophilic AD is highly promising and will be promoted in the future because of the lower amount of heat required for the mesophilic operation.

The C/N ratio is an important indicator for controlling the biological treatment system of AD. Most studies recommend an operating C/N ratio range of 20/1 to 30/1 with an optimal ratio of 25/1 for anaerobic bacterial growth in an AD system (Estevez et al., 2012). Improper C/N ratios can result in high total ammonia nitrogen (TAN) release and/or high volatile fatty acids (VFA) accumulation in the digester. Both TAN and VFAs are important intermediates and potential inhibitors in the AD process. The optimal C/N ratio varies with the type of feedstock to be digested. Yen and Brune (2007) used waste paper to balance the high nitrogen concentration of algal sludge for methane production and results showed that the optimized C/N ratio for the co-digestion was 20/1 to 25/1. Romano and Zhang (2008) recommended that the C/N ratio be maintained at 15/1 for the co-digestion of onion juice and digested sludge. However, the recent AD work concerning the C/N ratio was mostly focused on L-AD, and less information is available about the SS-AD.

The influence of operating parameters on AD is often studied separately, and contradictory results found in the literature maybe attributed to the interdependency (interaction) of these parameters. To our knowledge, the simultaneous effect of ISC, C/N ratio and temperature has not yet been investigated on SS-AD performance. However, due to the high sensitivity of dry media, their interaction may be critical for performing adequate SS-AD, and a strong interaction among these parameters may be present. The objective of the present study was to investigate the effects of three operating parameters, ISC, temperature and C/N ratio, and their interaction on the performance of SS-AD of RS under batch conditions. A Box–Behnken experimental design (BBD) combined with response surface methodology (RSM) was applied to determine the effect of these parameters on the biogasification. Microbial community structures in the SS-AD process were determined using the Illumina MiSeq platform.

2. Methods

2.1. Substrates and inoculum

Rice straw (RS) was collected from a rice field near the city of Chengdu (Sichuan province, China) and was cut into 30 mm-long pieces using a grinder. The anaerobic sludge used as inoculum was collected from an anaerobic digester in a sewage plant in Chengdu. The substrates and inoculum were individually homogenized and subsequently stored at 4 °C until further use. The chemical characteristic the RS and sludge are presented in Table 1.

2.2. Composting pretreatment

Because the components of lignocellulosic biomass (such as lignin, cellulose, and hemicellulose) are so difficult to degrade by bacteria-mediated hydrolysis that soluble compounds are less available for anaerobic microorganisms, the bioconversion efficiency is always limited (Taherzadeh and Karimi, 2008). Thus, the RS was firstly composted to facilitate the bioconversion efficiency. The composting method is described as follows: raw RS was put into a plastic drum, and 0.29 kg of inoculum and 1.57 kg of water were added into 1 kg of RS to obtain a moisture level of 65%; the mixture was homogenized, and then the drum was covered with plastic film to keep a temperature range of 60–70 °C. After the mixture was allowed to compost for 7 days, the chemical composition of the RS, including the TS, VS, lignin, cellulose, and hemicelluloses contents, were determined.

2.3. Solid-state anaerobic digestion (SS-AD)

The SS-AD experiment was performed in two sets. First, based on the results of the pretreatment experiment, an optimization experiment was conducted using the BBD method to evaluate the effect of ISC, temperature and C/N ratio, as well as their interactive effects, on the biogasification of composting RS. The experiment was conducted in self-designed anaerobic fermentation equipment. The equipment consisted of an anaerobic digester (2.5-L reagent bottle), a biogas collector (2-L Erlenmeyer flask), and a 1-L Erlenmeyer flask that collect the water discharged from the collector. The digesters were placed in a constant temperature incubator to maintain the experimental temperature. The biogas generated in the digester was transported into the head space of the bottle using a glass pipe. The water in the bottle was then pressed out and overflowed into the receiver through another glass pipe. The volume of the discharged water from the bottle represented the volume of the generated biogas in the digester. Based on the previous studies, three levels of ISC (20%, 27.5%, 35%), temperature (30 °C, 37.5 °C, 45 °C) and C/N ratio (15:1, 24:1, 34:1) were designed to evaluate their effect on the biogasification. A 550-g sample of composting RS was used as the digestion material,

Table 1
Characteristic of rice straw and inoculum for SS-AD.

Substrate	TS (%)	VS (% TS)	Total *C* (% TS)	Total *N* (% TS)	Lignin (% TS)	Cellulose (% TS)	Hemicellulose (% TS)
Rice straw	91.23 ± 0.41	79.87 ± 1.14	39.59 ± 0.57	0.66 ± 0.00	31.95 ± 0.66	30.22 ± 1.13	17.63 ± 0.04
Inoculum	9.80 ± 1.83	5.60 ± 0.12	36.85 ± 0.47	3.07 ± 0.14	/	/	/

The result are shown as means ± deviation ($n = 3$). TS, total solid; VS, volatile solid.

and the urea was added to adjust the C/N ratio. The amounts of substrate and urea at different levels are shown in Table 2. The fermentation lasted for 45 days.

BBDs are a class of rotatable or nearly rotatable second-order designs based on three-level incomplete factorial designs. The number of experiments (N) required for the development of a BBD is defined as $N = 2k\,(k-1) + C_0$, where k is the number of factors and where C_0 denotes the number of central points. Seventeen continuous-flow experiments were conducted based on the three levels of the ISC, temperature and C/N ratio. Response surface methodology (RSM) was used to build the best model and optimize the fermentation system using Design Expert version 6.06 (Stat Ease Inc., Minneapolis, MN, USA). The coded levels and corresponding real values of the operating variables considered in the experimental design are summarized in Table 2. Each treatment was performed in triplicate. One-way ANOVA was employed to study the primary effect sand interactions between the parameters selected on biogas production.

In the second experiment, a verification experiment was conducted to test the results from the optimization experiment. The experiment was conducted in a self-designed batch reactor with a working volume of 34 L. The reactor was 60 cm high and the radius was 15 cm. The batch reactor was made from organic glass, the roof, body and baseboard of which was connected with bolts and sealed with rubber rings. A water pipe and tailpipe was connected with the roof and a drain pipe was connected with the baseboard. Heat tape enwound the body of the reactor to maintain an appropriate reaction temperature, and a temperature detector was used to measure the interior-temperature through the hole of the body. The digestion experiment was operated for 45 days at the optimal conditions from the optimization experiment.

2.4. Analytical methods

The daily biogas production for each anaerobic digester was recorded using the water displacement method. The corresponding cumulative biogas volume was also calculated. The methane content in the produced biogas was analyzed with Gas Chromatography (Model 7890A, Agilent, USA). The TS, VS, and pH of the materials were determined using the standard water and wastewater examination methods (APHA, 1998). Total C and total N content were analyzed using the elemental analyzer (Vario EL Cube, Elementar Inc., Germany). The cellulose, hemicellulose, and lignin contents were analyzed based on the method described by Van Soest and Wine (1967). Briefly, this procedure is based on sequential extraction with neutral and acidic detergents, followed by a strong acid extraction. The different fractions were as follows: (i) the soluble fraction extracted by neutral detergent, (ii) hemicelluloses extracted by 2 M hydrochloric acid detergent, (iii) cellulose extracted by 72% sulfuric acid, and (iv) lignin, the difference after the acid extraction. The micro-area chemical and fiber structure was analyzed through electron microscopy (S300N, Hitachi, Japan) and Fourier Transform Infrared Spectroscopy (FTIR) measurements (Spectrum, Perkin-Elmer, USA).

2.5. High-throughput sequencing-based microbial community analysis

2.5.1. DNA extraction, 16S rRNA gene amplification, and sequencing

To analyze the structure of the microbial communities in the solid-state anaerobic digester, 0.5 g of sludge incubated for 1, 3, 7, 20, 30, and 45 days was used for DNA extraction. Total DNA was extracted according to the method described previously (Rademacher et al., 2012). For sequencing, the 16S rRNA genes were amplified with primers 515F (5′-GTGYCAGCMGCCGCGGTA-3′) and 909R (5′-CCCC GYCAATTCMTTTRAGT-3′ targeting the bacterial and archaeal V4-V5 region in the 16S rRNA gene) with a 10-mer barcode at the 5′ end of primer 515F (Tamaki et al., 2011). To minimize PCR bias, two PCR reactions were set up for each sample, and the PCR products in the replicate reactions were pooled. The details for the PCR and related procedures were described previously (Li et al., 2013). The amplicons from each sample were pooled with an equimolar concentration and

Table 2
Box–Behnken design for SS-AD of rice straw.

Number	Coded			Uncoded			Response				
	ISC	Temperature	C/N	ISC (%)	Temperature (°C)	C/N	Straw (g)	Inoculum (g)	Water (g)	Urea (g)	Biogas production (mL g VS^{-1})
1	0	0	0	27.5	37.5	24:1	550	190.2	20.6	10.2	341.3
2	−1	0	1	20	37.5	33:1	550	270.2	260.7	4.8	447.4
3	1	0	1	35	37.5	33:1	550	170.4	0	5.2	219.8
4	0	0	0	27.5	37.5	24:1	550	190.2	20.6	10.2	360.4
5	1	0	−1	35	37.5	15:1	550	170.4	0	21.4	185.3
6	−1	−1	0	20	30.0	24:1	550	270.2	260.7	9.9	278.7
7	−1	0	−1	20	37.5	15:1	550	270.2	260.7	21.2	344.4
8	0	1	1	27.5	45.0	33:1	550	190.2	20.6	5.1	61.7
9	1	1	0	35	45.0	24:1	550	170.4	0	10.3	58.0
10	−1	1	0	20	45.0	24:1	550	270.2	260.7	9.9	83.5
11	0	0	0	27.5	37.5	24:1	550	190.2	20.6	10.2	335.9
12	0	0	0	27.5	37.5	24:1	550	190.2	20.6	10.2	350.1
13	0	0	0	27.5	37.5	24:1	550	190.2	20.6	10.2	342.9
14	0	−1	−1	27.5	30.0	15:1	550	190.2	20.6	21.3	170.9
15	0	−1	1	27.5	30.0	33:1	550	190.2	20.6	5.1	168.7
16	0	1	−1	27.5	45.0	15:1	550	190.2	20.6	21.3	175.8
17	1	−1	0	35	30.0	24:1	550	170.4	0	10.3	148.1

ISC: initial solid concentration.

sequenced with the Illumina MiSeq platform (Illumina Company, San Diego, CA, USA).

2.5.2. Sequencing data processing

The raw sequences were sorted based on the unique sample barcodes, trimmed for sequence quality, and denoised using the QIIME pipeline (Caporaso et al., 2010). Chimera sequences were removed using the UCHIME algorithm (Edgar et al., 2011). As the sequence number varied among samples, we randomly resampled the sequences to 5600 reads per sample for further analysis. Then, the sequences were clustered by the complete linkage clustering method in the QIIME pipeline. Operational taxonomic units (OTUs) were classified using 97% identity of the 16S rRNA gene sequence as a cutoff, and the OTUs table was generated for each sample and used for statistical analysis. The Shannon diversity index and Chao1 estimator were calculated at 97% sequence identity in the Ribosomal Database Project pipeline (RDP) (http://pyro.cme.msu.edu/). The phylogenetic affiliation of each sequence was analyzed with the RDP classifier at a confidence level of 80%. To assure the accuracy of the RDP classifier results, the representative sequences of dominant bacteria and archaea were subjected to BLAST homology search against the non-environmental sequences and non-metagenomes in the NCBI nucleotide database (http://blast.ncbi.nlm.nih.gov).

2.6. Statistical treatment

Data are expressed as the means ± standard deviations of the triplicate measurements. Differences between mean values were examined by one-way analysis of variance, and statistical significance was assumed at a level of $P < 0.05$. The statistical analyses were performed using the software program SPSS 15.0 (SPSS Inc., Chicago, USA). The general differences of microbial community structure were evaluated by principal coordinates analysis (PCoA) in Fast UniFrac (http://bmf.colorado.Edu/fastunifrac/) using the relative abundances of the whole microbial communities.

3. Results and discussion

3.1. Effect of composting on the chemical composition of RS

Table 3 shows the composition of RS after composting, expressed in percentage of the initial TS. Compared with untreated RS, TS content of composting RS were significantly decreased by 63.6% ($P < 0.05$). A considerable amount of lignocellulose appeared to be decomposed and converted to other components. However, the decomposition rates were different; the hemicellulose content decreased by 7.5%, cellulose decreased by 64.5%, and lignin decreased by 13.6%. More cellulose was decomposed compared to lignin and hemicellulose. These results indicate that composting can effectively improve the biodegradability of complex substrates. Fdez-Guelfo et al. (2011) reported that pretreatment by composting resulted in a higher specific microbial growth rate (160–205%) compared to untreated organic fractions of OFMSW. Thus, the reason for the decreased lignocellulosic composition of RS may be due to the higher production of hydrolytic enzymes, which is induced by the increased specific microbial growth by composting. Contrasting with the result from Brummeler and Koster (1990), who found the pre-composting treatment of OFMSW resulted in a 19.5% volatile solid loss, our result shows that higher volatile solid contents were found in composting RS than those of the untreated sample, indicating that composting produced more available substances from RS that can be used by methanogen. Structural modifications of RS during composting treatment is supported by microscope observations (Supplementary material 1). The content of total *C* did not differ significantly between the untreated straw and the composting straw ($P > 0.05$), but this does not imply that composting has no effect on the *C* fraction of straw because the *C* fraction of the total content was only determined in this study. A portion of the unavailable *C* fraction may have been converted to soluble *C* content that can be used by bacteria. This hypothesis, however, requires further investigation and laboratory testing.

3.2. Modeling and optimization of SS-AD performance

3.2.1. 1Modeling of SS-AD for biogas production

Based on the aforementioned analysis, compositing was shown to exhibit favorable effectiveness on the degradation of the hemicellulose structure of RS; thus, composting is advised as a pretreatment method for SS-AD. BBD and RSM were used in 17 runs of the experiment to investigate the effect of three variables, ISC, temperature and C/N ratio, on the biogasification of SS-AD. The results in Table 2 were used for ANOVA and multiple regression analysis in Design-Expert software using the polynomial model equation. This approach enabled prediction of the optimum degree of biogas production and its corresponding optimum variables. One-way ANOVA was employed to test and analyze different models. The best-fit model found was a quadratic model for the biogasification of RS. The *F*-value of the model was 8.40 with a very low probability (*p*) value of 0.00523 (Table 4). This implies a highly significant effect on the biogas production process. However, the "lack of fit" having a *p*-value of 0.0006 showed significance, implying that the prediction for the fit of the model was not good. Thus, the model was adjusted to a better prediction as Eq. (1) (built with codified factors), where the regression coefficient is high ($R^2 = 0.92$). The final equation used to obtain the highest rate of biogas production was as follows:

$$\begin{aligned}\text{Biogas production} = {} & 295.92 - 58.00 \times A - 41.41 \times B + 2.26 \\ & \times C + 22.46 \times AB - 14.63 \times AC - 23.90 \\ & \times BC - 20.98 \times A^2 - 153.46 \times B^2 - 19.11 \times C^2 \quad (1)\end{aligned}$$

where *A* is the ISC, *B* is the temperature, and *C* is the C/N ratio. The results of the ridge analysis revealed that the optimal conditions were 20% (ISC), 35.6 °C (temperature), and 29.59 (C/N ratio), which results in a predicted biogas production of 346 mL g VS^{-1}, the theoretical maximum in Eq. (1). A good convergence between the experimental and simulated biogas production values was demonstrated (Supplementary material 2).

3.2.2. Main effects and interactions between parameters

The experimental results were visualized in three-dimensional response surface plots that show the simultaneous effect of two

Table 3
Effect of composting on the chemical composition of rice straw (% TS).

Substrate	TS	VS	Lignin	Cellulose	Hemicellulose	Total *C*	Total *N*
Rice straw	91.23 ± 7.41	79.87 ± 3.14	31.95 ± 2.66	30.22 ± 4.13	17.63 ± 0.33	39.59 ± 0.57	0.66 ± 0.07
Composting rice straw	33.21 ± 4.29	85.52 ± 1.44	27.6 ± 0.26	10.69 ± 1.14	16.30 ± 0.28	38.17 ± 1.01	0.78 ± 0.03

The result are shown as means ± deviation ($n = 3$). TS, total solid; VS, volatile solid.

Table 4
ANOVA for the response surface quadratic model.

Source	Statistics				
	Sum of squares	df	Mean square	*F*-value	Prob > *F*
Model	152073.3	9	16897.03	8.401935	0.0052
Residual	14077.62	7	2011.088		
Lack of fit	13815.25	3	4605.083	70.20839	0.0006
Pure error	262.3665	4	65.59163		
Corr total	166150.9	16			

factors on biogas production obtained from Eq. (1). Fig. 1a depicts a three dimensional plot showing the effects of ISC and temperature on the biogas production of composting RS. A large change in the biogas production by changing each parameter indicated that both parameters and their interaction are effective. The maximum value was achieved with the low amount of ISC (20%) and moderate temperature (35.6 °C). By increasing the ISC, the biogas production decreased drastically.

Fig. 2b shows that the increasing the temperature and C/N ratio to the optimal values, as determined by RSM, resulted in increased biogas production; however, further increases in concentration resulted in reduced biogas production. The optimal temperature and C/N ratio was 35.5 °C and 29.5. A lower C/N ratio causes ammonia accumulation and pH values exceeding 8.5, both of which are toxic to methanogenic bacteria. In contrast, a high C/N ratio is an indication of rapid consumption of nitrogen by methanogens and results in lower gas production (Verma, 2002). Previous report shows a wide range of optimal C/N ratios (10–90) in methane fermentation, but with 30 being the most common in liquid-digestion (Yen and Brune, 2007), consistent with our results.

The combined effect of ISC and C/N ratio on biogas production is shown in Fig. 2c. A strong negative interaction between ISC and

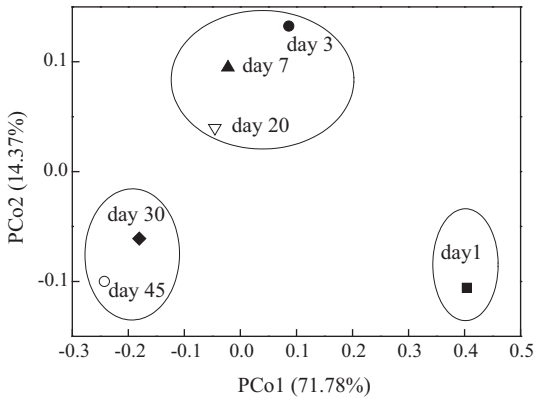


Fig. 2. The principal coordinates analysis (PCoA) based on the whole microbial communities at genus level in the SS-AD.

C/N was observed when the digestion temperature was fixed at 35 °C. The decreased ISC from 35% to 20% resulted in an increase in biogas production. This phenomena may explained by the bad performance of mass transfer, which lead the poor performance of microbial activity (Buffiere et al., 2005).

3.3. Results of verification experiment of SS-AD

3.3.1. Biogas production and methane content

To determine the validity of the predicted value, the experiments were performed at the optimal conditions. For ease of operation, the conditions were adjusted to ISC of 20%, 35 °C and C/N ratio of 30. The changes of biogas production and methane content during the SS-AD were shown in Supplementary material 3. The

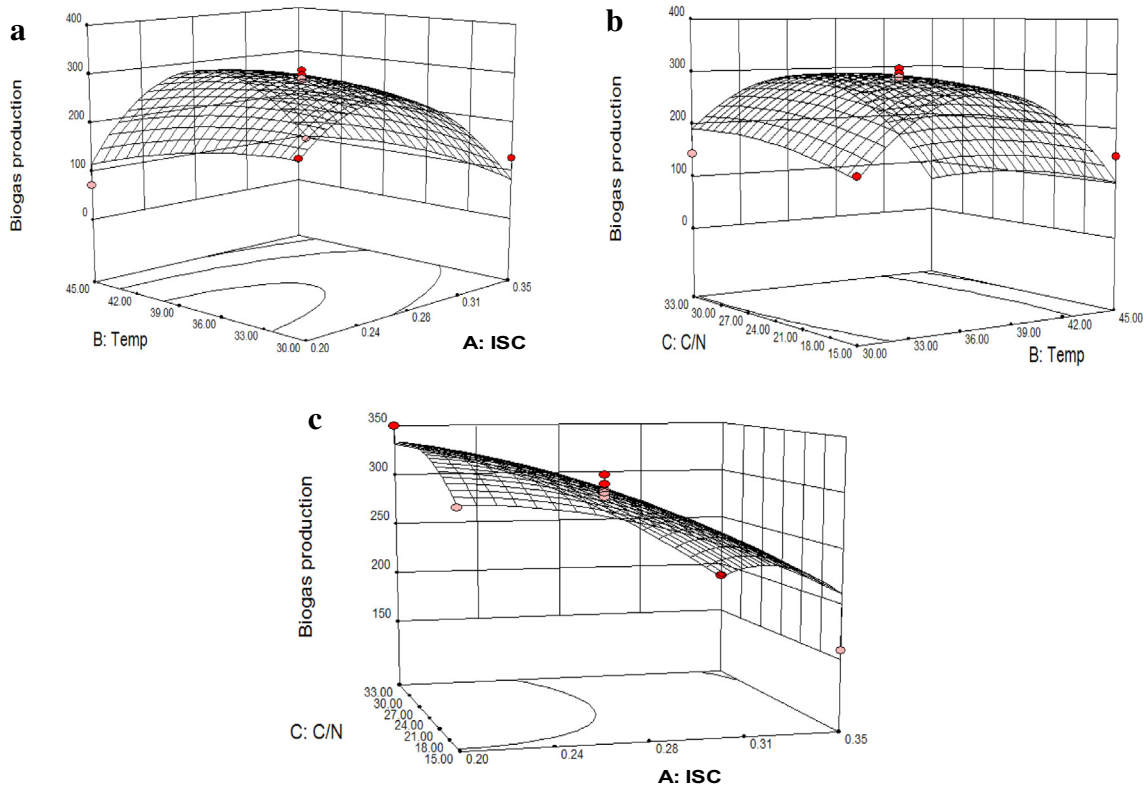


Fig. 1. Effects of ISC, temperature and C/N ratio on biogas production. (a) Interactive effect of ISC and temperature. (b) Interactive effect of temperature and C/N. (c) Interactive effect of ISC and C/N.

Table 5
The main absorbance bands in FTIR spectra and their assignments.

Band and peaks (cm^{-1})	Assignments
3600	Free OH of phenols or carbohydrates
3300–3400	H-bonded OH groups of alcohols, phenols and organic acids, as well as H-bonded NH groups
2920–2930	CH stretching of alkyl structures
1650–1630	Aromatic C═C, C═O in carboxylic acid, ketone and quinone
1600	Aromatic C═C, COO—, C═O
1509–1515	Aromatic skeletal vibrations
1100–1200	Amide III, —C—O—C— of carbohydrates, aromatic, Si—O—C
890–1000	Polysaccharides
898	β-(1 → 4)-Glycosidic linkages

Table 6
Microbial diversity indices based on 97% identity of 16S rRNA gene sequences and 3979 reads per sample.

Sample	Chao1 estimator of richness	Observed species	Shannon diversity index
Day 1	751.86	532	6.52
Day 3	900.88	624	7.41
Day 7	898.86	581	7.15
Day 20	957.94	645	7.71
Day 30	695.21	470	6.82
Day 45	866.88	530	6.89

experimental results yielded a biogas production of 353 mL g VS^{-1}, which was consistent with the result from the modeling of SS-AD (346 mL g VS^{-1}). Biogas production presented several small peaks before production finally ceased and the peak value was gradually decreased. This phenomenon may be related to the dynamic balance between the metabolism of acidogenic bacteria and methanogen in the AD (Qin et al., 2011). The highly concentrated substrate at the initial phase of AD supplied sufficient organic acid for the methanogen, which hastened the growth of methanogen and resulted in an increased biogas yield. The shortage of organic acid as it gradually decreased in the substrate limited the activity of methanogen, but stimulated acidogen, causing a drop in biogas yield. The activity of methanogen increased again when organic acid accumulated to an extent, resulting in the appearance of a peak in the biogas production. However, the peak value in the biogas production decreased because the concentration of the substrate is not as high as the initial concentration.

The methane production on the first day was low (only 1.69 mL g VS^{-1}) because the inoculum employed was not adapted to the substrate type and/or the operating conditions, i.e., temperature and dry state (Cho et al., 2013). As the microorganisms were gradually adapting to the AD environment (particularly, the heterogeneous nature of the substrate in AD systems might create an ideal micro-environment for the growth of the methanogen population), methane production greatly increased linearly to 4.82 mL g VS^{-1} at day 7, subsequently decreased to 2.8 mL g VS^{-1} at day 12 and also increased to 4.82 mL g VS^{-1} at day 27 and gradually decreased thereafter. The dynamics of methane production could be closely related to the methanogen population, and this hypothesis was investigated thoroughly in the following microbial community Section 3.3.3.

3.3.2. Morphological characteristics of RS

The morphological changes that were induced by SS-AD were examined by SEM and FT-IR spectra to obtain insight into the structural modifications of RS during SS-AD (Supplementary material 4), the structural loosening of cells with a simultaneous increase in porosity was found in the digested RS, and partial exposure of cellulose networks, similar to a honeycomb, was found in the straw after SS-AD was removed. The FT-IR spectra were analyzed for the RS before and during SS-AD, and the assignments of the primary recorded bands are shown in Table 5. The digested RS showed apparent differences with the undigested straw (Supplementary material 5). The intensities of absorption at 3300–3600 cm^{-1} and 1600–1650 cm^{-1} were decreased with an increase of digestion time, indicating that macromolecular organic matter such as alcohols, phenols, carbohydrates and organic acids, were degraded gradually. In all of the spectra, aromatic skeletal vibrations related to the lignin content were assigned at 1509–1515 cm^{-1} and 1100–1200 cm^{-1}. The bands in the 1200–890 cm^{-1} region represent the polysaccharides, and the band at 898 cm^{-1} represents β-(1 → 4) glycosidic linkages. Both of these two bands were related to the decomposition of cellulose and hemicelluloses. These band intensities of digested samples were lower than those of undigested samples. The FTIR analysis showed that the changes included a decrease in the association of lignin with carbohydrates, and some degradation and solubilization of cellulose and hemicelluloses occurred during the process of AD. These structural changes in the substrate indicated a successful performance of the SS-AD because the decomposition of these substances means the utilization of substrate by the microorganisms in the system.

3.3.3. Microbial community analysis by high-throughput sequencing

To reveal the microbial community structure in the SS-AD, digestion sludge samples retrieved from the reactor incubated for 1, 3, 7, 20, 30, and 45 days were used to obtain DNA for the amplification and subsequent sequencing of a region of archaea and bacterial 16S rRNA genes. A total of 129,340 high-quality sequences were obtained (ranging from 5600 to 10,446 per sample, data not shown); all of the sequences were aligned and clustered to calculate operational taxonomic units (OTUs) using 97% sequence identity as a cutoff, resulting in 459–840 OTUs at a sequencing depth of 3979 reads per sample. Clearly, the Chao1 estimator of richness, observed species, and Shannon diversity index of the microbial community at day 3, day 7, and day 20 were higher than those at day 1, day 30, and day 45, and the highest value was found at day 20 (Table 6). These results indicated a dynamic change of microbial communities during the process of SS-AD. The microbial community seems more complicated in the growing period than that in the start-up and stabilization period. The differences in microbial communities between different fermentation stages were also demonstrated by principal coordinates analysis (PCoA). This analysis showed a distinct community structure between six samples (Fig. 2) that can be separated into three groups. The samples from day 3, day 7, and day 20 were clustered together, the samples from day 30 and 45 were clustered together, and the sample from 1th day was clustered alone.

Fig. 3 shows the microbial community composition at class level in the process of SS-AD. Clearly, the microbial community changed as the AD proceeded. *Methanobacteria*, *Bacteroidia*, *Clostridia*, *Betaproteobacteria*, and *Gammaproteobacteria* are the primary prokaryotic communities in the SS-AD. *Methanobacteria*, the primary community of methanogens belonging to *Euryarchaeota* of archaea, increased continuously to the 6.6% in the first 7 days, decreased slightly and then increased throughout the next days. *Methanobacteria* composed 1.16% of the community at day 30 and decreased thereafter. This change was consistent with the dynamics of methane production. A previous study showed that archaea accounted for up to 20% in anaerobic digesters (Zakrzewski et al., 2012). In this study, although the primer set should cover most methanogens, the highest relative abundance of *Euryarchaeota* only accounted for less than 10% of total reads. One explanation might be the different microbial performance in

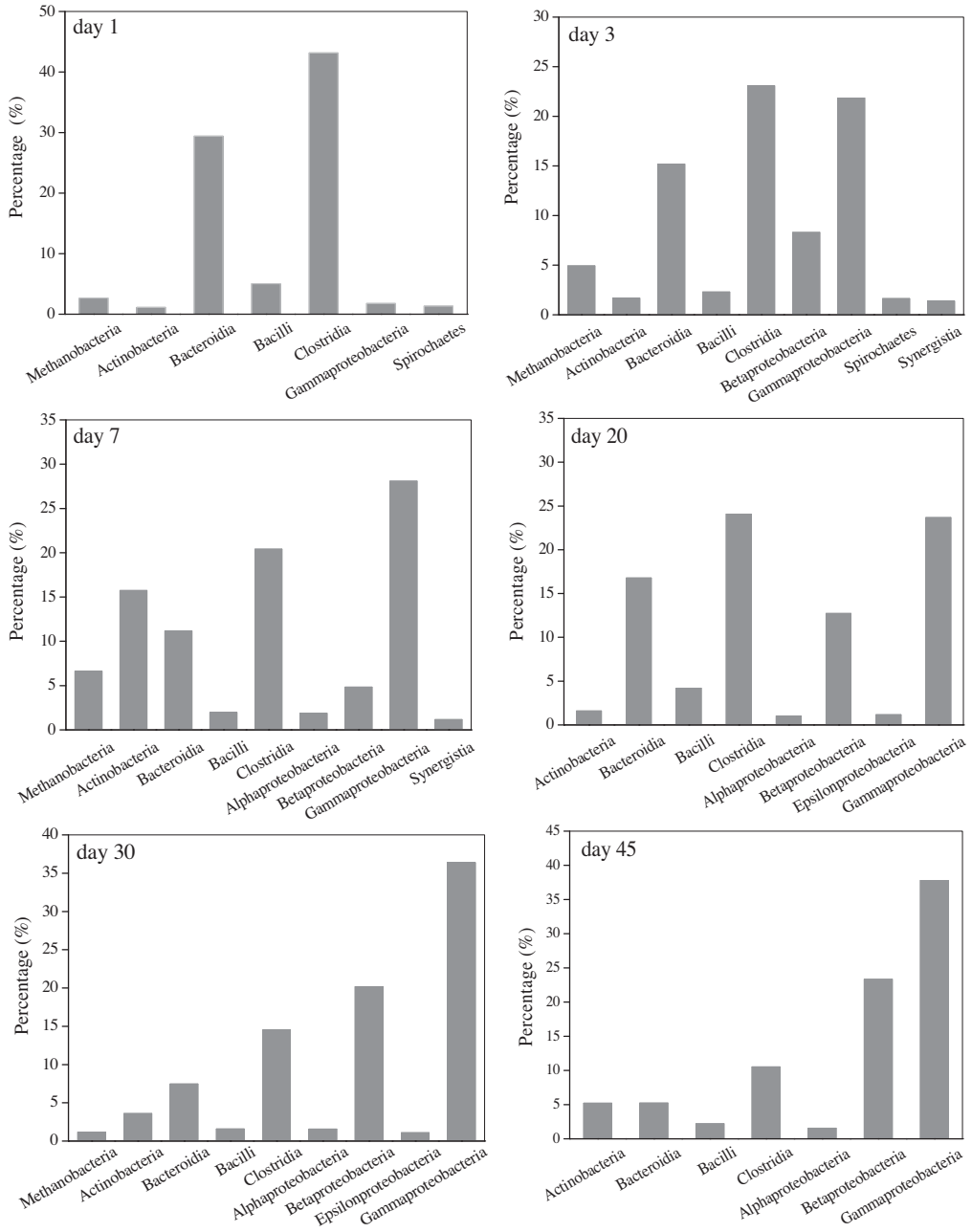


Fig. 3. Microbial community structure at class level in the SS-AD. Only the communities accounting for above 1% of the total community were shown.

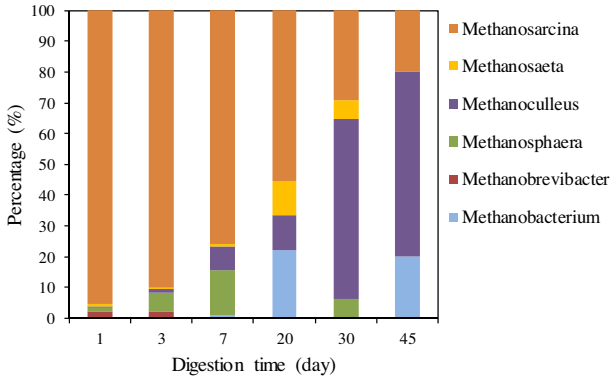


Fig. 4. Composition of methanogens at genus level in the SS-AD.

the different AD systems because the substrate used by Zakrzewski was the mixture of maize silage and chicken manure and because the digestion system used was liquid AD.

To reflect the community dynamics of the digestion process, the composition of methanogens at a genus level was further investigated (Fig. 4). Six genera of methanogens, *Methanosarcina*, *Methanosaeta*, *Methanoculleus*, *Methanosphaera*, *Methanobrevibacter*, and *Methanobacterium*, were found in the SS-AD. Certain studies showed that a hydrogenotrophic pathway prevailed in the digesters of agricultural waste (Jaenicke et al., 2011). Acetoclastic methanogenesis may be a major pathway in certain systems, or both pathways were utilized for methanogenesis (Li et al., 2013). In this system, acetoclastic methanogens, *Methanosarcina*, and hydrogenotrophic methanogens, *Methanoculleus*, accounted for on average 86.5% and 32.1%, respectively, indicating both pathways for methane production were most likely co-occurring during the

mesophilic fermentation of RS. Chu et al. (2010) reported that acetoclastic methanogens accounted for 74% of bacteria in an anaerobic digester of food waste under mesophilic conditions, and they detected 72% of clones affiliated with *Methanosaeta* and 2% of clones affiliated with *Methanosarcina* among 50 clones of methanogens, whereas the present study found little *Methanosaeta* (less than 12%) and abundant *Methanosarcina*. This discrepancy could be related to the different microbial reaction for different AD substrate. Additionally, in the present study, the methanogen community was overrepresented by the *Methanosarcina* at the start-up and growing periods (1–20 days) but decreased gradually as fermentation proceeded, whereas *Methanoculleus* gradually increased and was abundant at the stabilization period (20–45 days). The shifting from *Methanosarcina* to *Methanoculleus* during the process suggested the acetoclastic methanogens, although important in the initial phase of the reactor start-up and growing period, were displaced by hydrogenotrophic methanogens at the subsequent stabilization period.

4. Conclusion

Composting pretreatment can effectively improve the degradation of lignocellulosic structure of RS. The optimization experiment showed a significant interactive effect of temperature, ISC and C/N ratio on SS-AD. The RSM results indicated the optimal conditions of 35.6 °C, 20% ISC and a C/N ratio of 29.6:1 for biogasification. *Methanobacteria*, *Bacteroidia*, *Clostridia*, *Betaproteobacteria*, and *Gammaproteobacteria* were the primary microbial communities in SS-AD of composting RS. The dominant methanogen was shifted from *Methanobacterium* to *Methanoculleus* during the SS-AD process. The experimental results serve as useful information for understanding the AD system.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.biortech.2014.11.089.

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